Ali Saei

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Summary

- AI/ML engineer for wireless systems — O-RAN digital twins, 5G/6G channel modeling (NVIDIA Sionna, Wireless InSite), and RAN optimization. Built deep learning models with 7000$\times$ faster inference than ray-tracing (30s $\rightarrow$ 4ms, <4% error), validated via field measurements. Best Demonstration Award, NYU Wireless Brooklyn 6G Summit; demo at the AI-RAN Alliance booth, MWC 2026.
- Strong in deep learning (CNNs, transformers, diffusion models), reinforcement learning, and real-time GPU/FPGA systems. Built real-time 5G testbeds (OpenAirInterface, Open5GS, USRP X410) and a GPU-accelerated channel simulator reaching 95% of OTA throughput at sub-100µs latency. Fluent in PyTorch, TensorFlow, ONNX, and TensorRT.

Skills

- **Programming:** Python, C, C++, CUDA, Verilog, MATLAB
- **AI/ML Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn, ONNX, TensorRT
- **AI Tooling & Agents:** Claude Code, Agentic Workflows, LLM Automation
- **Wireless & SDR Tools:** OpenAirInterface (OAI), srsRAN, Open5GS, GNU Radio, RFNoC, NVIDIA Sionna/pyAerial/ARC, Wireless InSite, RF Instruments
- **Systems & DevOps:** Docker, OpenShift, Linux, Git, Bash, Slurm, NVIDIA Nsight
- **Data Processing:** NumPy, Pandas, Matplotlib, OpenCV

Experience

Apple — Cupertino, CA
Software Engineer Intern – Connectivity, Cellular & Satellite (CCS) — June 2026 – Aug 2026

Research Experience

Research Assistant — WiNES Lab, Northeastern University
Advisor: Prof. Tommaso Melodia (Sep. 2022 – Present)

- Built deep learning channel models for 5G/6G testbeds trained on 1M+ scenarios, deployed on Colosseum (world's largest channel emulator) and Sionna SYS for real-time wireless Digital Twins; benchmarked against Wireless InSite and NVIDIA Sionna RT and validated with 3TB+ of field measurements across Boston.
- Built a GPU-accelerated (CUDA) real-time channel simulator for OpenAirInterface, replacing CPU-based RF-Sim/vrtsim and reaching sub-100µs slot processing; added lock-free shared-memory IPC with master-timeline sync for multi-UE 5G digital-twin experiments on an Open5GS core.
- Developed the first multi-UE channel emulator on USRP X410 — a 4-channel full-duplex FPGA (RFNoC) system at 1µs latency, validated with commercial Sierra modems and an OAI gNB.

Research Assistant — Telecommunications Innovation Lab (TIL), University of Tehran
Advisor: Prof. Maryam Sabbaghian (Aug. 2021 – Aug. 2022)

- Developed a Multi-Agent RL (MARL) trajectory-optimization framework for mmWave Aerial Base Stations, with a custom mmWave channel simulator (LoS/NLoS detection) benchmarking MADDPG, PPO, and SAC.

Education

Northeastern University — Boston, MA
PhD Candidate in Electrical Engineering (Overall GPA: 4.0 / 4.0) — Sep 2022 – Present
MS in Electrical Engineering (Overall GPA: 3.9 / 4.0) — Sep 2022 – May 2026

University of Tehran — Tehran, Iran
BS in Electrical Engineering (Communication Systems – Major GPA: 3.81 / 4.0) — May 2022

Publications

Journal Articles

AIRMap: AI-Generated Radio Maps for Wireless Digital Twins.
Saeizadeh, A., Tehrani-Moayyed, M., Villa, D., Beattie, J. G., Johari, P., Basagni, S., & Melodia, T. **IEEE Transactions on Wireless Communications**. **Best Demonstration Award**, NYU Wireless Brooklyn 6G Summit. (2025)

Conference Papers

AI-Assisted Agile Propagation Modeling for Real-Time Digital Twin Wireless Networks.
Saeizadeh, A., Tehrani-Moayyed, M., Villa, D., Beattie, J. G., Wong, I. C., Johari, P., Anderson, E. W., Basagni, S., & Melodia, T. In *Proc. IEEE International Workshop on Computer Aided Modeling and Design of Communication Links and Networks (CAMAD)*. (2024)
Demonstrations: 6G Symposium (Washington D.C.), IEEE ICC (Denver, CO), and Northeastern University Industry Day.

A Multi-Modal Non-Invasive Deep Learning Framework for Progressive Prediction of Seizures.
Saeizadeh, A., Schonholtz, D., Neimat, J. S., Johari, P., & Melodia, T. In *Proc. IEEE 20th International Conference on Body Sensor Networks (BSN)*. (2024)

SeizNet: An AI-Enabled Implantable Sensor Network System for Seizure Prediction.
Saeizadeh, A., Schonholtz, D., Uvaydov, D., Guida, R., Demirors, E., Johari, P., Jimenez, J. M., Neimat, J. S., & Melodia, T. In *Proc. 19th Wireless On-Demand Network Systems and Services Conference (WONS)*. (2024)

Preprints & Under Review

GENESIS: Harnessing AI Agents for Autonomous 6G RAN Synthesis, Research, and Testing.
Aghayev, T., Elkael, M., Polese, M., Nguyen, M. D., Gemmi, G., Lacava, A., **Saeizadeh, A.**, Prasad, R., Testolina, P., Feraudo, A., Nanda, S., Johari, P., D'Oro, S., & Melodia, T. *arXiv preprint arXiv:2605.27360*. (2026)

Patents

Systems and Methods for Predicting an Onset of a Neurological Event in a Human or Animal.
Johari, P., **Saeizadeh, A.**, Melodia, T., & Jimenez, J. H. *U.S. Patent Application No. 19/094,041*. (2025)

Awards & Recognition

- **iContest, Apple (2026)** – 2nd Place
- **NYU Wireless Brooklyn 6G Summit (2025)** – Best Demo Award for "AIRMap: AI-Generated Radio Maps for Wireless Digital Twins"
- **7th Annual College of Engineering PhD Research Expo (2025)** – Audience Choice Award
- **NSF I-Corps Entrepreneurship Award (2024)** – $50K grant for technology commercialization
- **IEEE Body Sensor Networks (BSN) Conference (2024)** – IEEE Student Travel Grant

Leadership & Innovation

- **NSF I-Corps Entrepreneur (2024):** Selected (<10% acceptance) to commercialize seizure prediction technology. Led 100+ customer interviews, developed business model canvas, validated $2.3B addressable market.

Professional Service and Teaching

- **Peer Reviewer (2023–2026)** IEEE Transactions on Wireless Communications, INFOCOM, Globecom, WCNC, PIMRC, CCNC, and IEEE Access, among others.
- **Teaching Assistant / Head TA (2020–2022)** Signal Processing, Signals & Systems, Communication Systems, and Intelligent Systems.